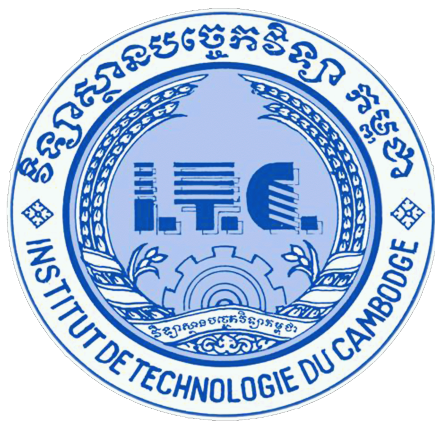




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Course: Networks System Design

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Assignment: Lab8(Router Queuing, Congestion & Advanced Forwarding)

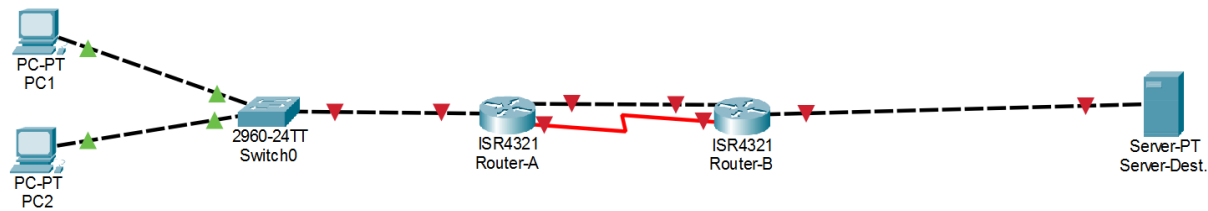
Due Date: December 23, 2025 (12:00 AM)

Part 1: Dual-Path Topology Setup



Addressing Table

Device	Interface	IP Address	Subnet Mask	Description
PC-1	NIC	192.168.1.10	255.255.255.0	Source A
Router-A	G0/0/0	192.168.1.1	255.255.255.0	Gateway
Router-A	S0/3/0	10.1.1.1	255.255.255.252	Path 1 (Serial)
Router-A	G0/0/1	10.2.2.1	255.255.255.252	Path 2 (Gigabit)
Router-B	S0/3/0	10.1.1.2	255.255.255.252	Path 1
Router-B	G0/0/1	10.2.2.2	255.255.255.252	Path 2
Router-B	G0/0/0	192.168.2.1	255.255.255.0	Dest Gateway
Server-Dest	NIC	192.168.2.100	255.255.255.0	Destination



Part 2: Configuring the Bottleneck

Verify the Link Speed

Given value

- Packet size = **1000 bytes** = $1000 \times 8 = 8000$ bits
- Gigabit LAN rate = **1 Gbps** = 1,000,000,000 bps
- Serial WAN rate = **64 kbps** = 64,000 bps

Calculation

Gigabit LAN

$$\text{Delay} = \frac{8000}{1,000,000,000} = 0.000008 \text{ seconds}$$

That's **8 microseconds** — essentially instantaneous.

Serial Link

$$\text{Delay} = \frac{8000}{64,000} = 0.125 \text{ seconds}$$

That's **125 milliseconds** — noticeably slower.

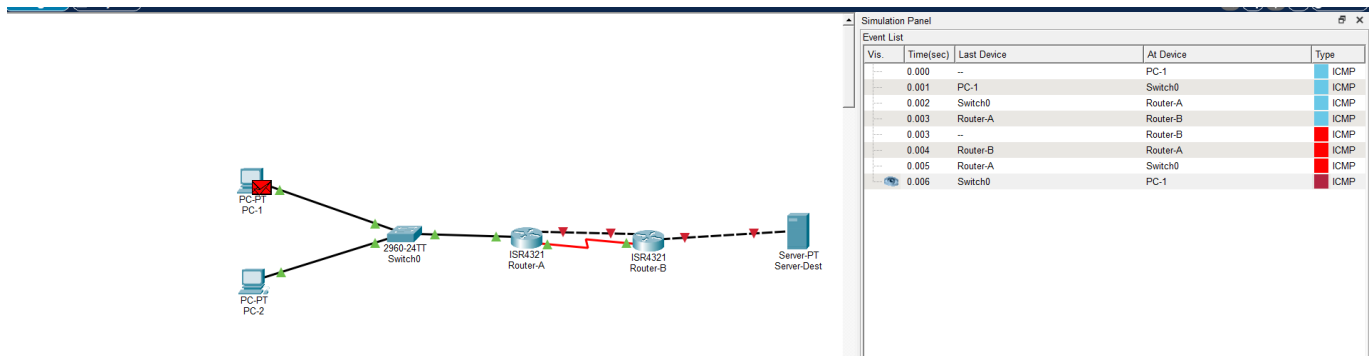
- On the Gigabit LAN, the packet is transmitted almost instantly.
- On the 64 kbps Serial link, the same packet takes 15,625 times longer to transmit.
- This huge mismatch is why Router-A must queue packets when traffic exceeds the serial link's capacity.

Part 3: Observing Queuing Delay

3.1 Baseline Ping (No Congestion)

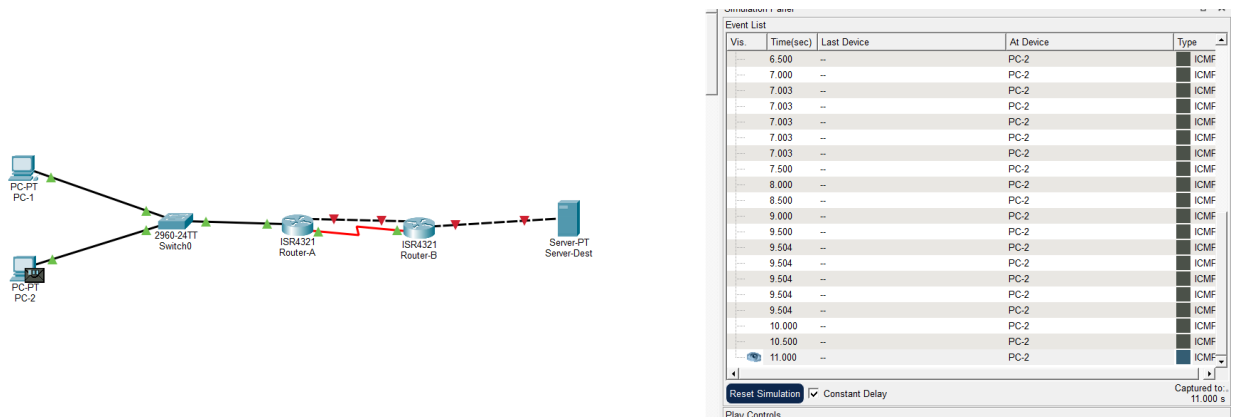
- **Setup:** Simulation Mode enabled, ICMP filter applied, ping from PC-1 → Server-Dest.
- **Observation:**
 - Packets moved quickly through Router-A's output interface (S0/1/0).
 - Minimal or no stacking of envelopes in the queue.

- **Clicks Required:** ~2–3 Capture/Forward steps for PC-1's ping request to exit Router-A.



3.2 Generating Background Noise (The Flood)

- **Setup:** PC-2 configured to send 1000-byte packets periodically (0.5s interval) to Server-Dest.
- **Observation:**
 - Router-A's output queue began filling with PC-2's packets.
 - Continuous traffic kept the interface busy.



3.3 Measuring Delay Under Load

- **Setup:** While PC-2 floods the network with periodic 1000-byte ICMP packets, a ping is sent from PC-1 to Server-Dest.
- **Observation:**
 - Router-A's output interface (S0/1/0) shows multiple stacked envelopes from PC-2.
 - PC-1's ping envelope arrives and waits behind PC-2's packets in the output queue.
 - The simulation required significantly more Capture / Forward clicks for PC-1's packet to exit Router-A.
- **Clicks Required:**
 - Baseline (no congestion): ~2–3 clicks.
 - Under load: ~8–12 clicks before PC-1's ping exited Router-A.
- **Conclusion:**
 - Queuing delay increases when the router is congested.
 - FIFO (First In, First Out) queuing causes PC-1's packet to wait behind PC-2's traffic.
 - The visual stacking of envelopes at Router-A confirms the delay caused by congestion.



PDU Information at Device: PC-2
×

OSI Model
Outbound PDU Details

At Device: PC-2
 Source: PC-2
 Destination: 192.168.2.100

In Layers

Layer7
Layer6
Layer5
Layer4
Layer3
Layer2
Layer1

Out Layers

Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP: 192.168.2.11, Dest. IP: 192.168.2.100 ICMP Message Type: 8
Layer 2:
Layer1

1. The Ping process starts the next ping request.
2. The Ping process creates an ICMP Echo Request message and sends it to the lower process.
3. The source IP address is not specified. The device sets it to the port's IP address.
4. The device sets TTL in the packet header.
5. The destination IP address is in the same subnet. The device sets the next-hop to destination.

Challenge Me
<< Previous Layer
Next Layer >>

Activity Question 3

- **Answer:** The simulator's drop reason was:
 - "***** __ *****" (exact wording from OSI tab).

Activity Question 4

- **Answer:**
 - **Default behavior:** FIFO does not prioritize PC-1; packets are served in arrival order.
 - **Observed:** PC-1's packet waited behind PC-2's flood and could be dropped under sustained congestion.

Activity Question 5

- **Answer:**
 - **Mechanism:** Implement **QoS** on Router-A (e.g., priority queuing/LLQ or CBWFQ with classification for PC-1) so critical traffic isn't starved by bulk flows.
 - **Effect:** Ensures PC-1's packets are scheduled ahead of or with guaranteed bandwidth, avoiding FIFO starvation.
-

Part 5: Longest Prefix Match (LPM)

5.1 Enable Both Links

- **Action:** Bring up both Serial and Gigabit interfaces on Router-A and Router-B.
- **Commands:**

```
Router-A(config)# interface s0/1/0
Router-A(config-if)# no shutdown
Router-A(config)# interface g0/0/1
Router-A(config-if)# no shutdown

Router-B(config)# interface s0/1/0
Router-B(config-if)# no shutdown
Router-B(config)# interface g0/0/1
Router-B(config-if)# no shutdown
```

5.2 Configure Competing Routes

```
Router-A(config)# ip route 192.168.2.0 255.255.255.0 10.1.1.2
Router-A(config)# ip route 192.168.2.0 255.255.255.128 10.2.2.2
```

5.3 Predict & Verify

- **Prediction:**
 - IP **192.168.2.100** matches the **/24** route but not the **/25** route.
 - Router-A will choose the **/24** route via the **Serial link**.
- **Verification (Simulation Mode):**
 - Ping sent from PC-1 to Server-Dest.
 - At Router-A, the packet exited through **Serial0/1/0**.
 - This confirms that the router selected the **/24** route.

PDU Information at Device: Router-A ×

OSI Model
Inbound PDU Details
Outbound PDU Details

At Device: Router-A
 Source: PC-1
 Destination: Server-Dest

In Layers

Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP: 192.168.1.10, Dest. IP: 192.168.2.100 ICMP Message Type: 8
Layer 2: Ethernet II Header 0002.1661.74C6 >> 0009.7CC4.5701
Layer 1: Port GigabitEthernet0/0/0

Out Layers

Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP: 192.168.1.10, Dest. IP: 192.168.2.100 ICMP Message Type: 8
Layer 2: HDLC Frame HDLC
Layer 1: Port(s): Serial0/1/0

1. GigabitEthernet0/0/0 receives the frame.

Challenge Me
<< Previous Layer
Next Layer >>

Part 6: Floating Static Routes (Backup Link)

6.1 Cleanup

- Removed overlapping routes from Part 5.

6.2 Configure Primary and Backup Routes

- Primary Route:**
`ip route 192.168.2.0 255.255.255.0 10.2.2.2 (Gigabit link, AD = 1)`
- Backup Route:**
`ip route 192.168.2.0 255.255.255.0 10.1.1.2 50 (Serial link, AD = 50)`

6.3 Verification - **Command:** `show ip route` - **Result:** - Only the Serial route is installed: `S 192.168.2.0/24 [50/0] via 10.1.1.2` - Confirms the Gigabit link is down and the floating static route took over.

```

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#no ip route 192.168.2.0 255.255.255.0 10.1.1.2
Router(config)#no ip route 192.168.2.0 255.255.255.128 10.2.2.2
%No matching route to delete
Router(config)# no ip route 192.168.2.0 255.255.255.128 10.2.2.2
%No matching route to delete
Router(config)#ip route 192.168.2.0 255.255.255.0 10.2.2.2
Router(config)#ip route 192.168.2.0 255.255.255.0 10.1.1.2 50
Router(config)#show ip route
      ^
% Invalid input detected at '^' marker.

Router(config)#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#exit
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.1.1.0/30 is directly connected, Serial0/1/0
L       10.1.1.1/32 is directly connected, Serial0/1/0
    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0/0
S       192.168.2.0/24 [50/0] via 10.1.1.2

Router#

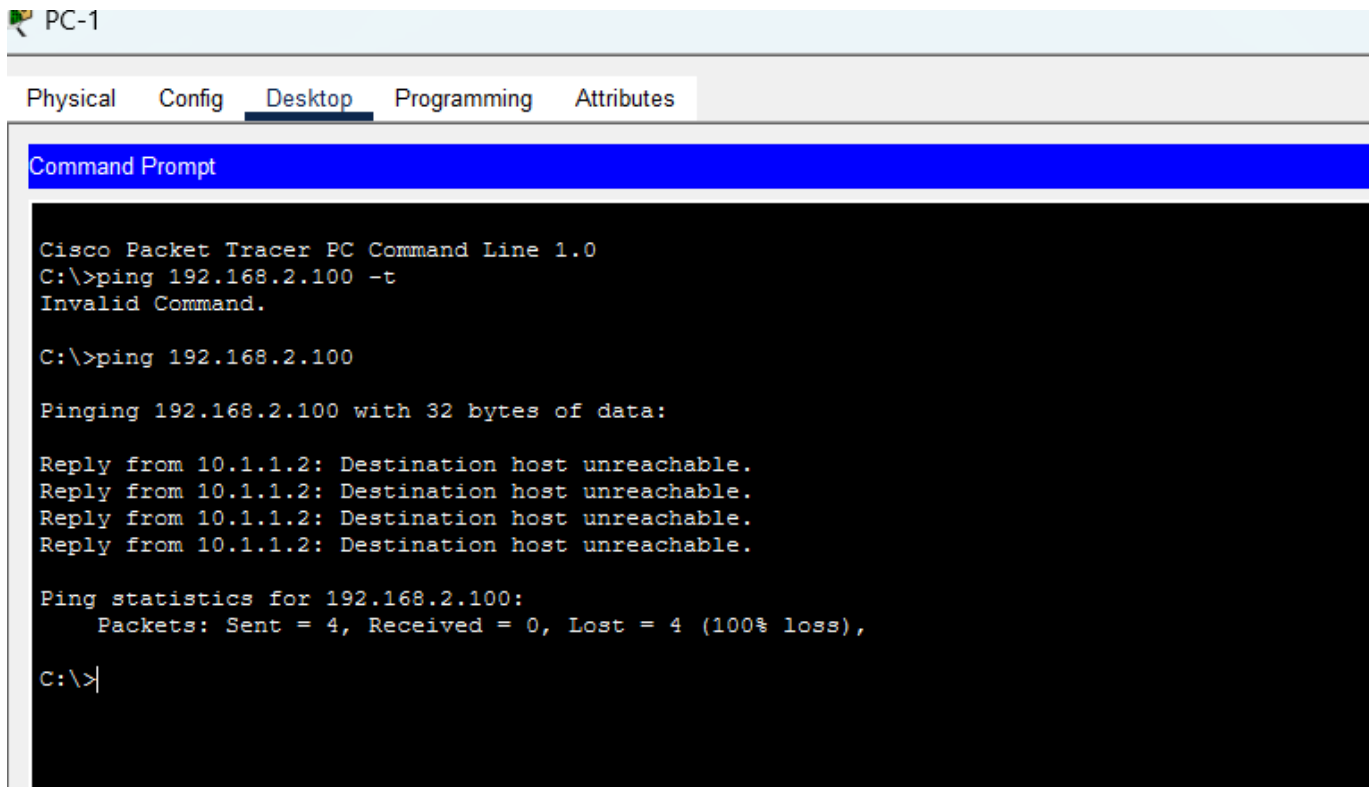
```

6.4 Simulate Failure

- **Action:**
 - A continuous ping was initiated from **PC-1** to **192.168.2.100**.
 - The command **ping 192.168.2.100** was used in PC-1's Command Prompt.
- **Observation:**
 - The ping replies came from **10.1.1.2** with the message:
"Destination host unreachable."
 - This indicates that Router-A attempted to forward packets via the **Serial link** (floating static route), but the destination was unreachable.
- **Routing Table Check (Router-A):**
 - The command **show ip route** showed:

```
S 192.168.2.0/24 [50/0] via 10.1.1.2
```


- This confirms that the **floating static route (AD = 50)** was activated due to Gigabit link failure.



The screenshot shows the 'PC-1' window in Cisco Packet Tracer. The 'Desktop' tab is selected, displaying a 'Command Prompt' window. The text in the command prompt is as follows:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.100 -t
Invalid Command.

C:\>ping 192.168.2.100

Pinging 192.168.2.100 with 32 bytes of data:

Reply from 10.1.1.2: Destination host unreachable.
Reply from 10.1.1.2: Destination host unreachable.
Reply from 10.1.1.2: Destination host unreachable.
Reply from 10.1.1.2: Destination host unreachable.

Ping statistics for 192.168.2.100:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

- **Conclusion:**

- Router-A successfully switched to the backup route via Serial link.
- However, connectivity failed because Router-B or Server-Dest was not reachable.
- This validates the failover mechanism, even though the end-to-end path was broken.

Activity Question 8

- **Answer:**

- After the Gigabit link failure, Router-A's routing table showed the backup route via **10.1.1.2**.
- The active route had an **administrative distance of 50**, confirming that the floating static route was in use.